

Product Specification — SettleIQ

1. Product

SettleIQ — AI-powered settlement reconciliation and exception resolution

One-line description

SettleIQ helps finance teams reconcile payment and settlement data, investigate financial exceptions using evidence-backed AI, automatically resolve high-confidence cases, and escalate uncertain cases for human review.

2. Problem

Finance teams receive financial data from different sources. Clean records can be matched using deterministic rules, but exceptions require manual investigation.

Typical questions include:

- Why does the settlement amount differ from the payment amount?
- Why is a payment missing from the settlement?
- Is a transaction duplicated?
- Are two records referring to the same transaction despite different references?
- Can this discrepancy safely be resolved, or does it need human review?

The objective is therefore **not simply to match records**, but to reduce the manual effort required to investigate and resolve exceptions.

3. Target User

Primary user

Finance / Finance Operations team

Specifically, someone responsible for:

- payment reconciliation
- settlement reconciliation
- investigating discrepancies
- resolving exceptions
- reviewing unresolved financial records

We are building for the **finance operator**, not the end customer.

4. Input Data

For the MVP, SettleIQ will work with three synthetic datasets:

payments.csv

Contains payment/transaction information.

settlements.csv

Contains settlement information associated with payments.

fees.csv

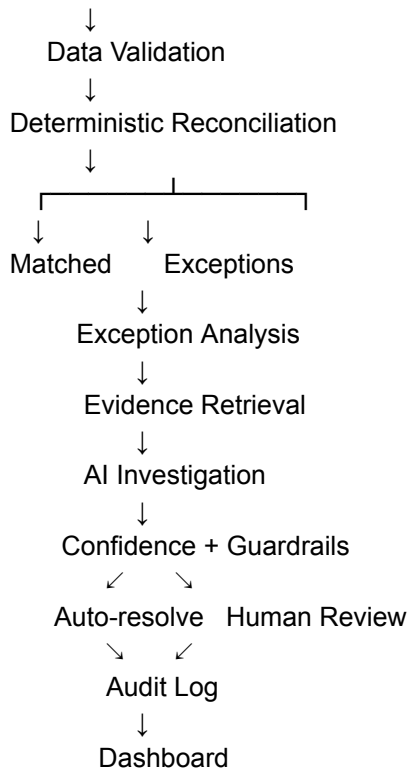
Contains applicable processing/platform fees.

No refunds in MVP.

We deliberately keep the initial data model small so that the core reconciliation workflow can be completed reliably.

5. Core Workflow

Payment + Settlement + Fee Data



This is the **core product loop**.

6. Reconciliation Engine

The first layer should be deterministic.

It will attempt to match payment and settlement records using known attributes such as:

- transaction/payment ID
- settlement reference
- amount
- relevant dates
- transaction status

We do **not** need an LLM for straightforward matching.

The system should first establish:

Matched
Exception

This separation is important because it demonstrates proper AI judgment.

7. MVP Exception Types

We'll initially support **five categories**:

1. Amount mismatch

Payment and settlement amounts differ.

Example:

Payment: ₹5,000
Settlement: ₹4,850
Fee: ₹150

Potential explanation: processing fee.

2. Missing settlement

A payment exists but no corresponding settlement can be found.

3. Duplicate transaction

Multiple records appear to represent the same underlying transaction.

4. Reference mismatch

The payment and settlement references don't directly match, but related information may indicate that they belong together.

5. Unknown / ambiguous exception

The system cannot establish a sufficiently reliable explanation.

These cases should generally be routed toward **human review rather than forced automation**.

8. Role of AI

AI enters **after deterministic reconciliation identifies an exception**.

Its job is to:

1. Investigate the exception.
2. Gather relevant evidence.
3. Determine the likely root cause.
4. Explain the discrepancy.
5. Recommend an action.
6. Assign a confidence score.
7. Identify the evidence supporting the recommendation.

For example:

Exception:

₹250 settlement discrepancy

Evidence:

Payment = ₹5,000

Settlement = ₹4,750

Fee record = ₹250

AI conclusion:

Difference is consistent with recorded processing fee.

Confidence:

96%

Recommendation:

AUTO_RESOLVE

9. What AI is NOT allowed to do

This is a critical product rule.

The LLM **does not directly modify financial records or autonomously make unrestricted financial decisions.**

The AI produces a recommendation.

A deterministic **guardrail layer** decides whether that recommendation is permitted.

For example:

AI says:
AUTO_RESOLVE
Confidence: 96%

↓

Guardrail checks:

- ✓ Evidence exists
- ✓ Evidence is relevant
- ✓ Confidence > threshold
- ✓ Known resolution rule satisfied

↓

AUTO-RESOLVE

If the conditions aren't satisfied:

→ HUMAN REVIEW

This gives us **bounded AI autonomy**.

10. Successful Resolution

An exception is considered successfully auto-resolved when:

- the system has sufficient supporting evidence,
- the AI's confidence exceeds the configured threshold,
- the proposed resolution satisfies deterministic guardrails,
- the resolution matches the known ground truth in our test dataset.

This last point is particularly important.

We will create a synthetic dataset with **known ground truth**, allowing us to objectively evaluate whether the system actually made the correct decisions.

11. Human Review

Cases that cannot safely be resolved automatically enter a review queue.

The finance operator can:

- inspect the evidence
- see the AI explanation
- approve a recommendation
- reject it
- mark the case for further investigation

Every human decision is recorded.

12. Audit Trail

Every investigated exception should maintain:

Transaction ID
Exception type
Evidence used
AI conclusion
Confidence
Guardrail result
Final action
Human action (if applicable)
Timestamp

This allows us to answer:

"Why did the system make this decision?"

That is especially important for a financial operations product.

13. Dashboard

The MVP dashboard should provide a simple overview:

SETTLEMENT RECONCILIATION

Records processed	200
Matched	164
Exceptions	36

Auto-resolved	24
Human review	12

Match rate	82%
Auto-resolution rate	66.7%

These values will be generated from actual test runs.

We will **never manufacture performance numbers for the pitch.**

14. Settlement Q&A — Advanced Feature

Once the core system works, we'll add the differentiating layer.

The finance operator can ask questions such as:

"Why is today's settlement ₹68,600 lower than expected?"

SettleIQ can investigate the underlying reconciled data and respond with an evidence-backed breakdown.

For example:

Settlement shortfall: ₹68,600

₹31,200 → processing fees

₹18,400 → unresolved refunds [future]

₹11,000 → pending settlements

₹8,000 → unresolved exceptions

For our current MVP, since refunds aren't included, the actual breakdown will only use the data sources we support.

The important part is:

The Q&A layer sits on top of our reconciliation system.

It is **not** a separate generic chatbot.

15. Success Metrics

We'll evaluate the system using the synthetic ground-truth dataset.

Primary metrics:

Reconciliation accuracy

How many records are correctly classified as matched/exception.

Exception classification accuracy

How accurately we identify the type of exception.

Auto-resolution accuracy

Of the cases we automatically resolve, how many are actually correct?

Auto-resolution coverage

What percentage of exceptions can be safely resolved without human intervention?

False auto-resolution rate

How often does the system incorrectly resolve something automatically?

This is particularly important.

We would rather escalate an uncertain case than incorrectly resolve a financial discrepancy.

16. MVP Boundary

MUST HAVE

- Payment dataset
- Settlement dataset
- Fee dataset
- Data validation
- Deterministic reconciliation
- Five exception types
- Evidence retrieval
- AI investigation
- Structured AI response
- Confidence score
- Guardrail engine
- Auto-resolution
- Human review queue
- Audit trail
- Dashboard
- Evaluation metrics

SHOULD HAVE

- Evidence-linked explanations
- Better exception prioritization
- Polished investigation interface
- Larger synthetic dataset
- Failure/recovery handling

STRETCH

- Settlement Q&A
- Natural-language financial investigation
- Advanced analytics
- Exception trends
- What-if analysis

17. Explicitly Out of Scope

For the first version:

- Real Razorpay API
- Real bank APIs
- Real financial transactions
- Real customer data
- Refund processing
- Tax reconciliation
- Cash forecasting
- Payment execution
- Fully autonomous financial decisions
- Multi-agent architecture

This prevents scope creep.

18. Product Success Criterion

At the end of the build, we should be able to demonstrate:

A finance operator uploads a realistic batch of payment, settlement and fee records. SettleIQ reconciles them, identifies exceptions, investigates those exceptions using relevant evidence, correctly resolves high-confidence cases, sends uncertain cases to human review, records every decision, and produces measurable reconciliation results.

If we can do **that extremely well**, we have a legitimate Razorpay Buildathon submission.